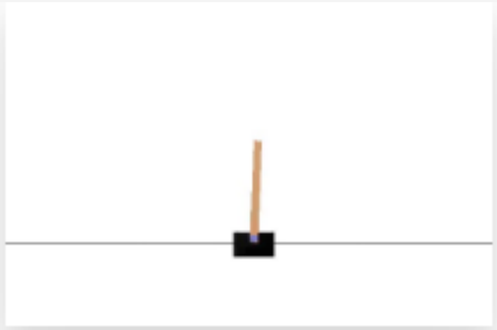
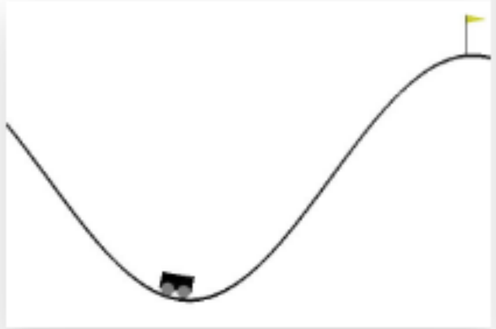
Librerías con Entornos de RL

Manuel Sánchez-Montañés

OpenAI Gym



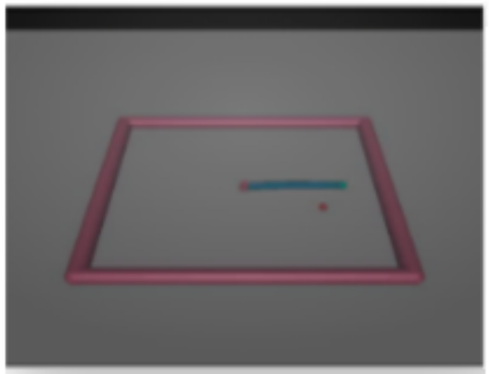
CartPole-v0
Balance a pole on a cart (for a short time).



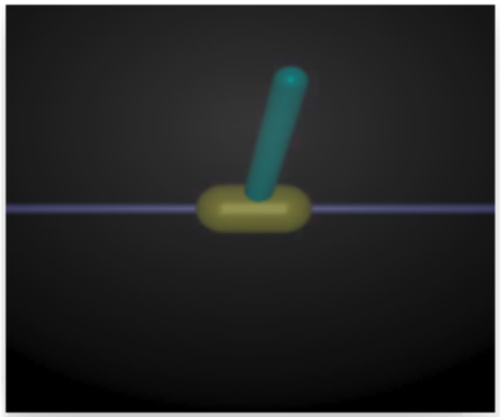
MountainCar-v0
Drive up a big hill.



Pendulum-v0
Swing up a pendulum.



Reacher-v2
Make a 2D robot reach to a randomly located target.



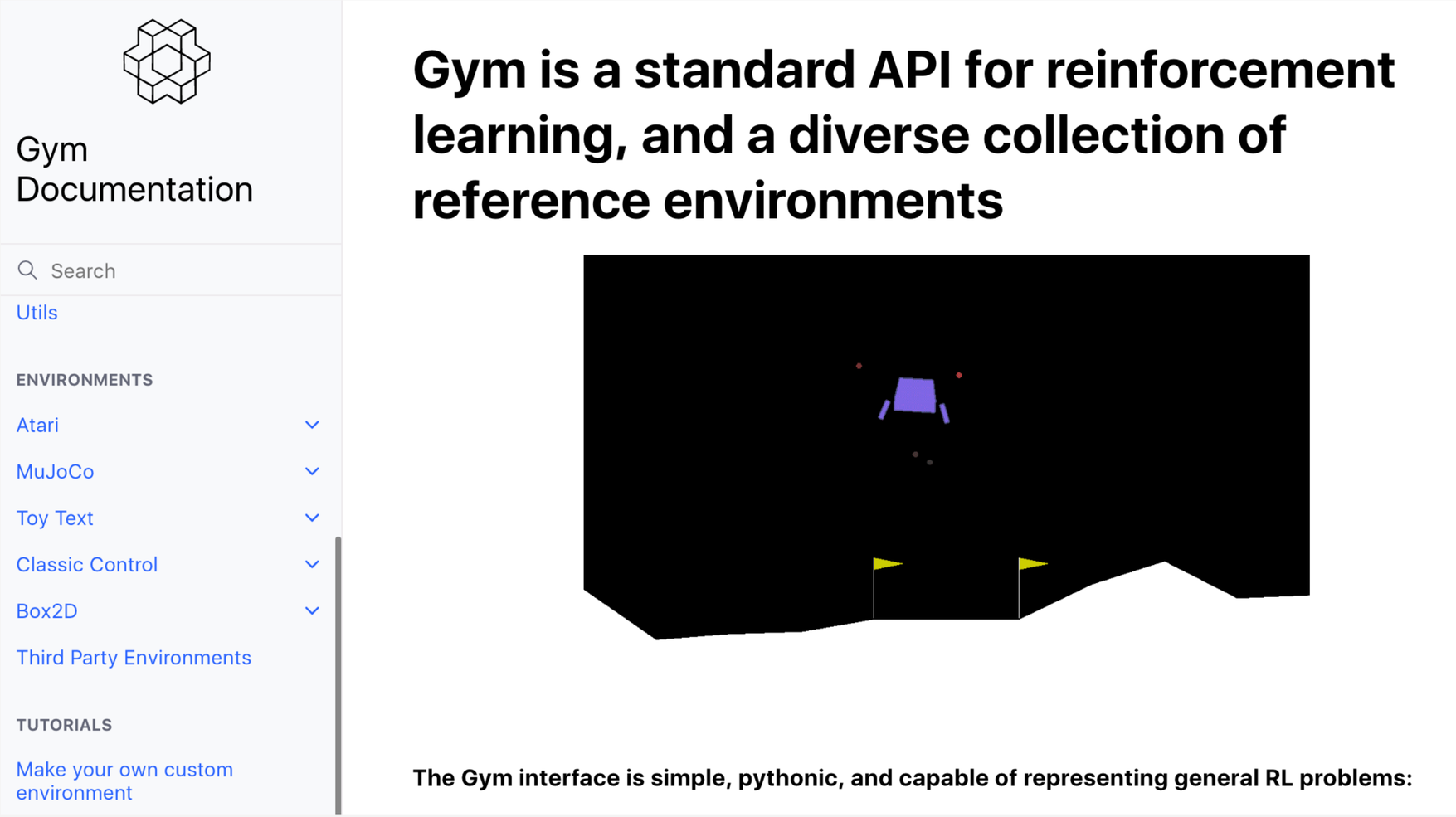
InvertedPendulum-v2
Balance a pole on a cart.



MsPacman-ram-v0

OpenAI Gym

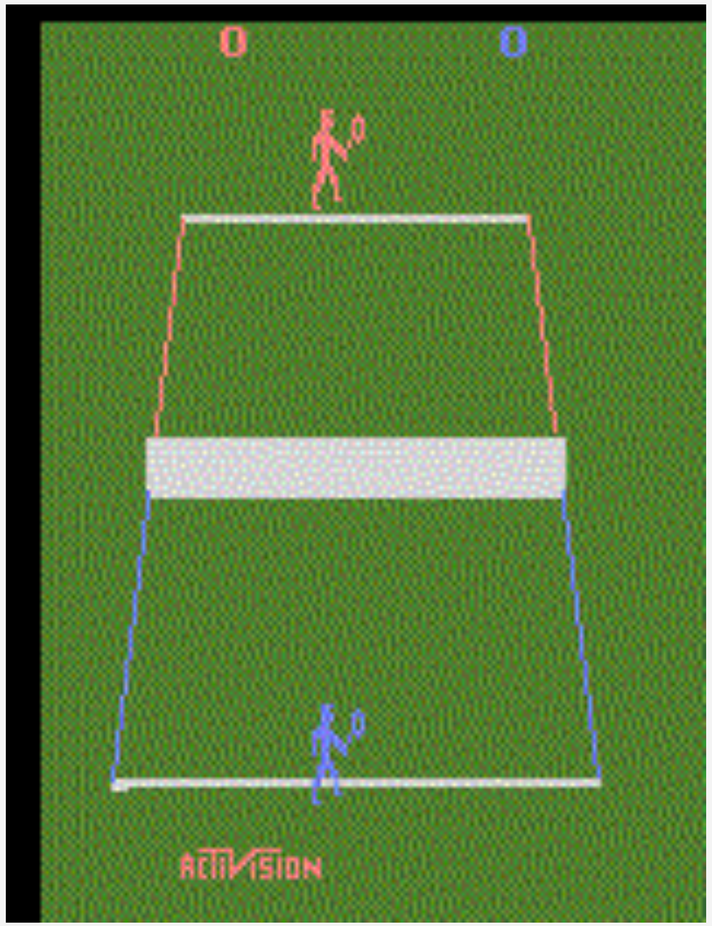
https://www.gymlibrary.dev/



Gym: Atari 2600 Arcade



Gym: Atari 2600 Arcade



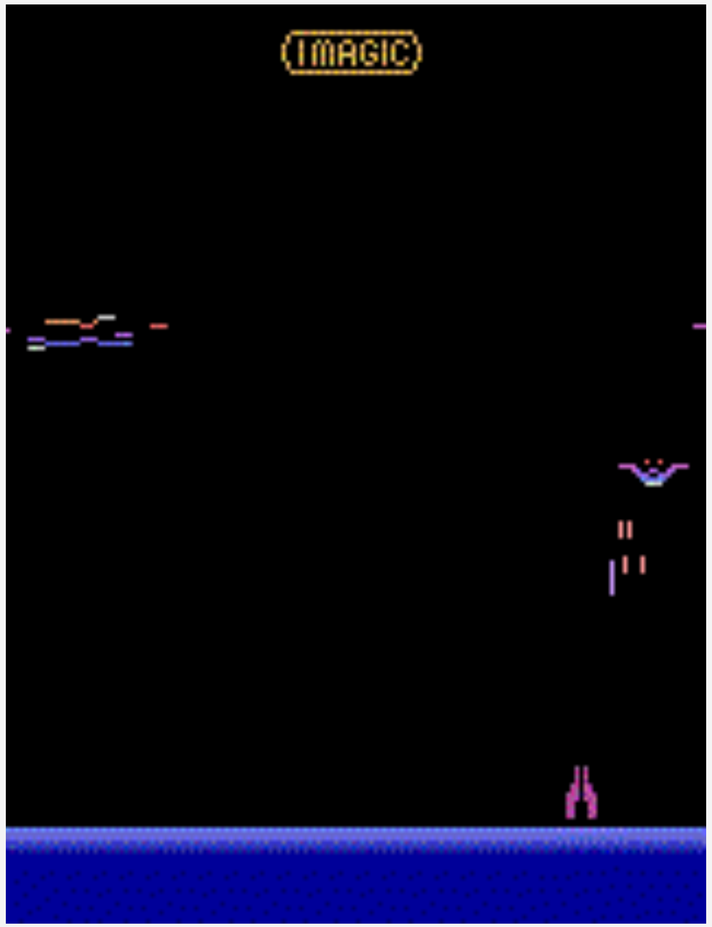
Tennis

Gym: Atari 2600 Arcade



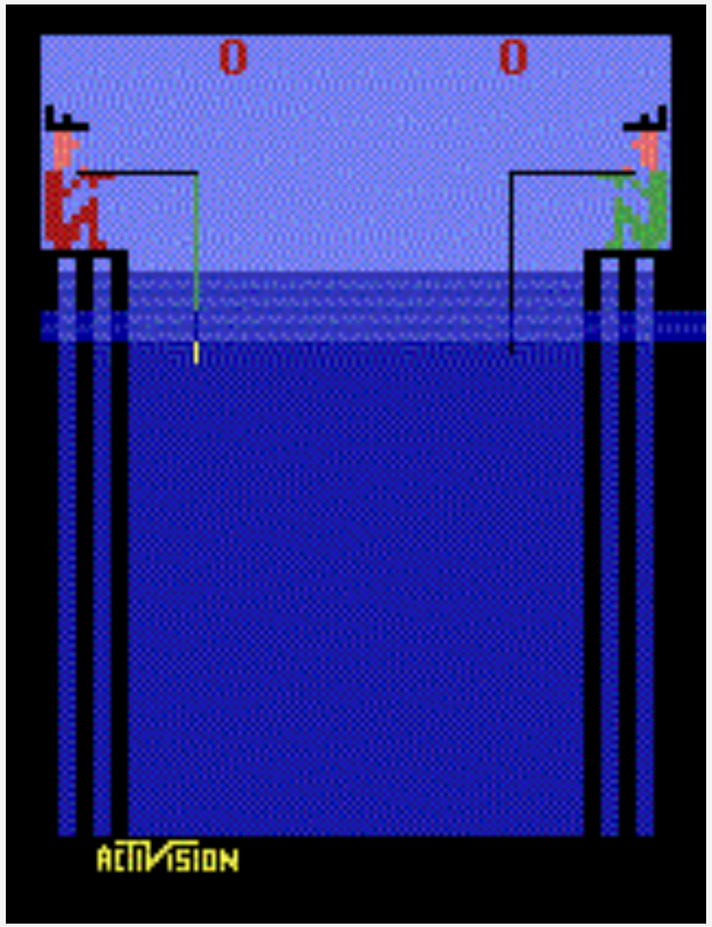
Skiing

Gym: Atari 2600 Arcade



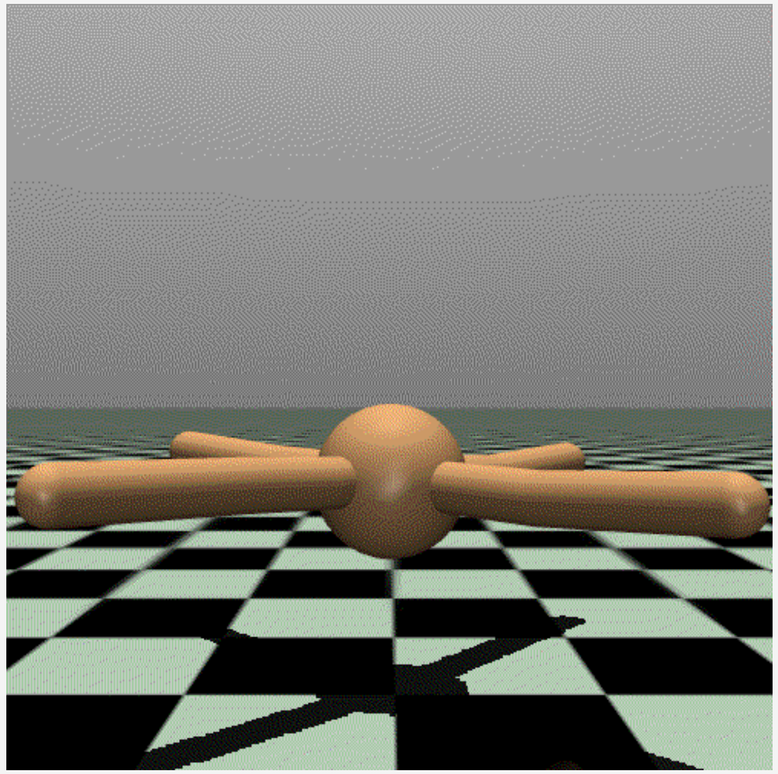
Demon attack

Gym: Atari 2600 Arcade

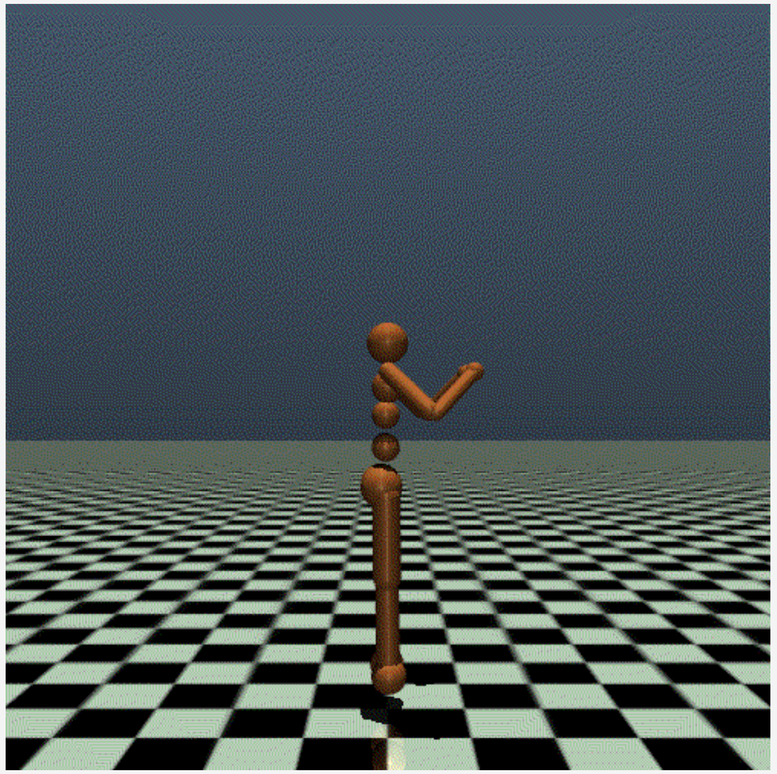


Fishing derby

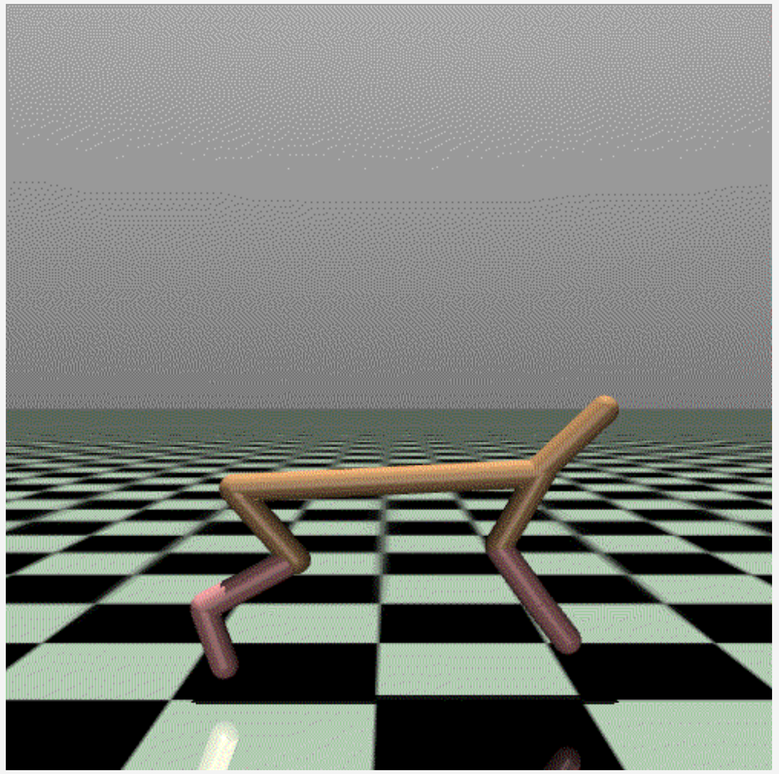
Gym: Mujoco (Multi-joint dynamics in contact)



Ant

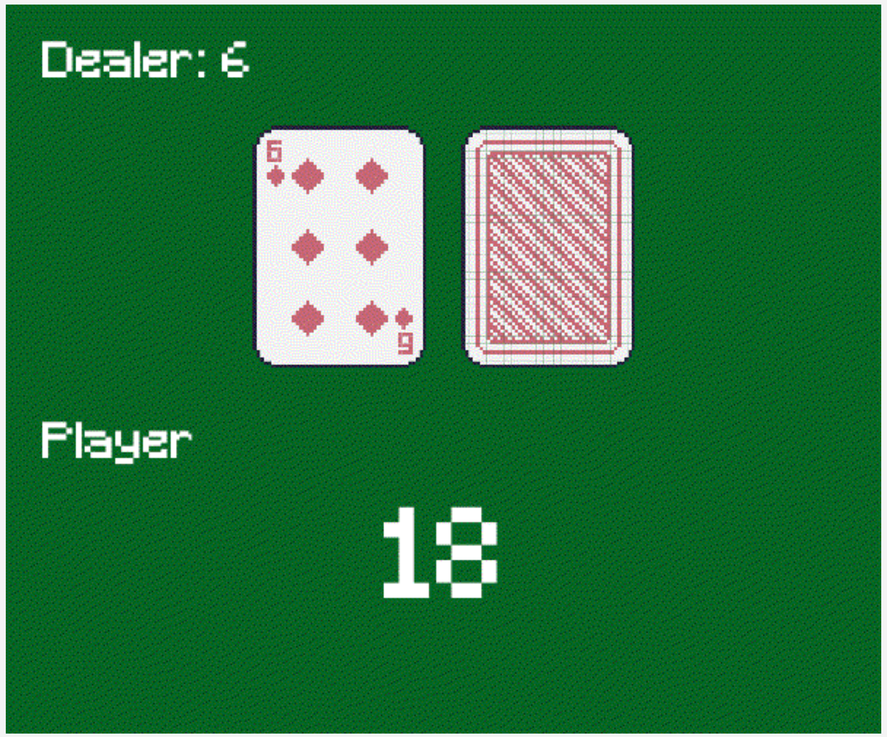


Humanoid

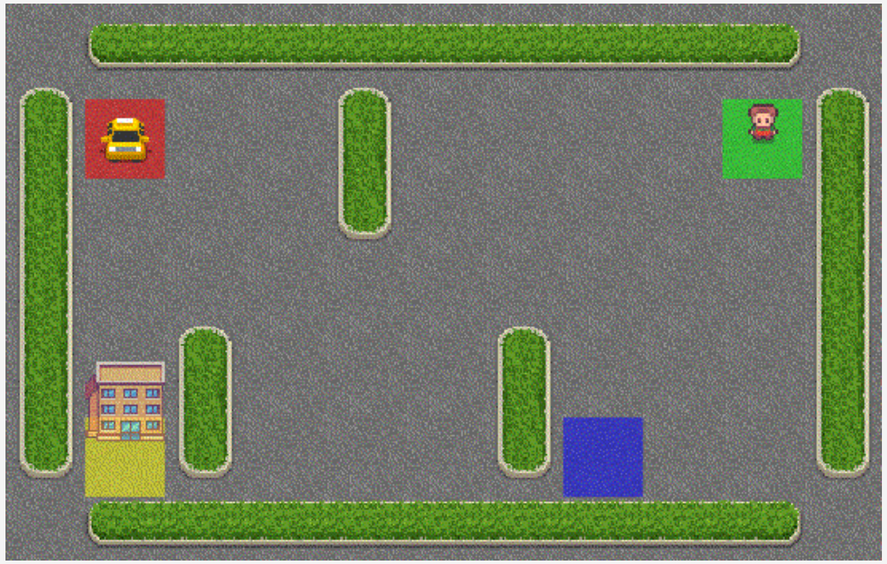


Half Cheetah

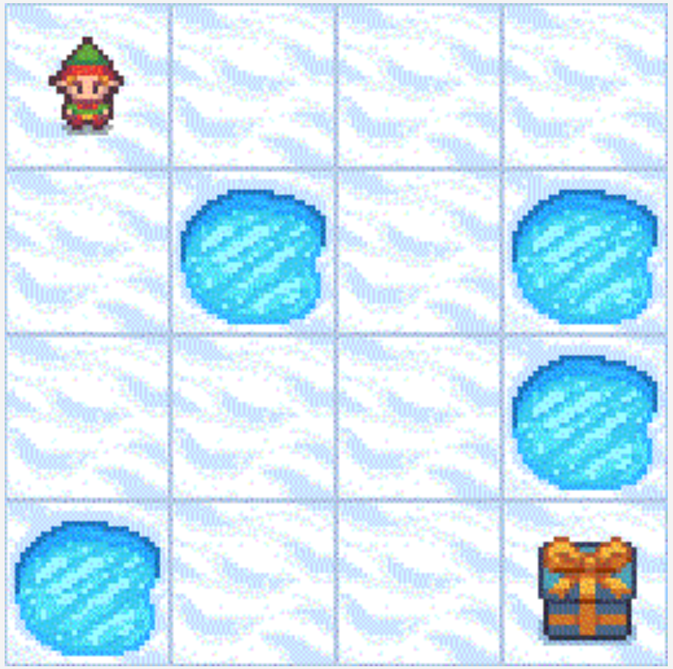
Gym: Toy Text



Blackjack

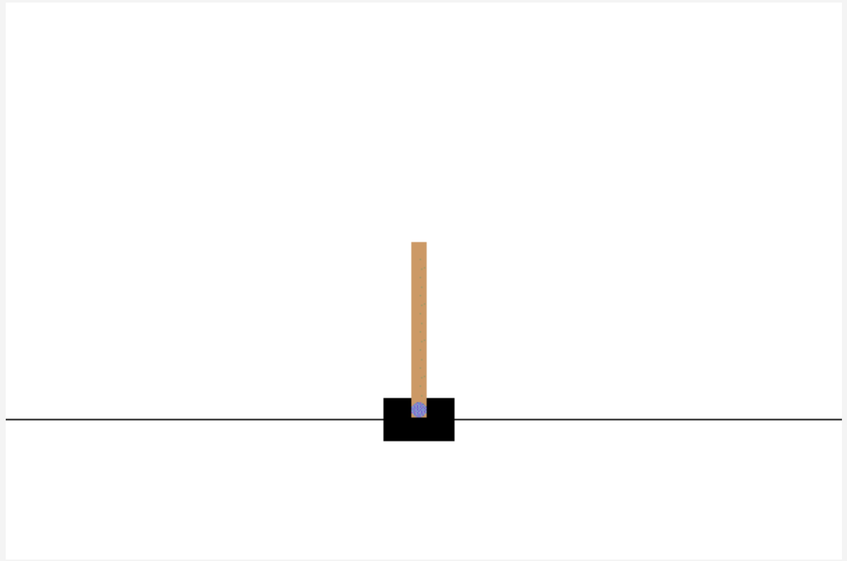


Taxi

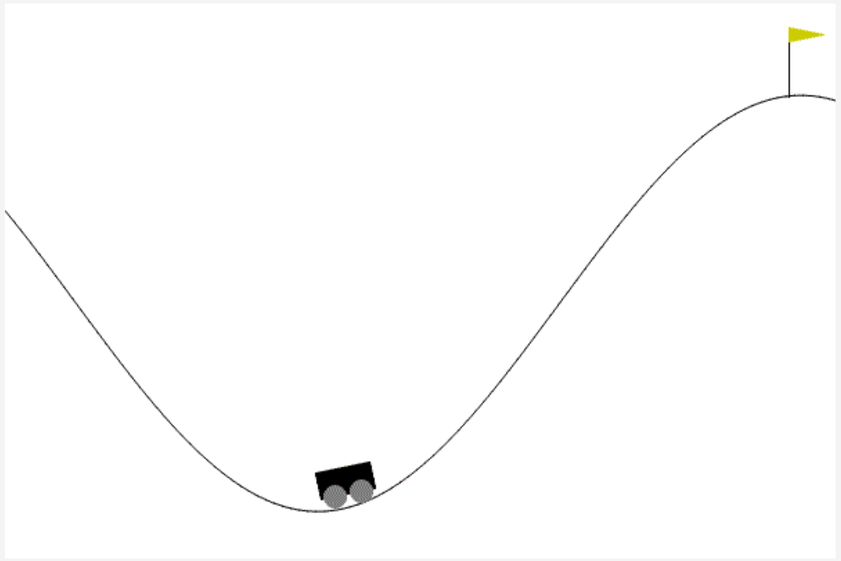


Frozen lake

Gym: Classic control



Cart pole



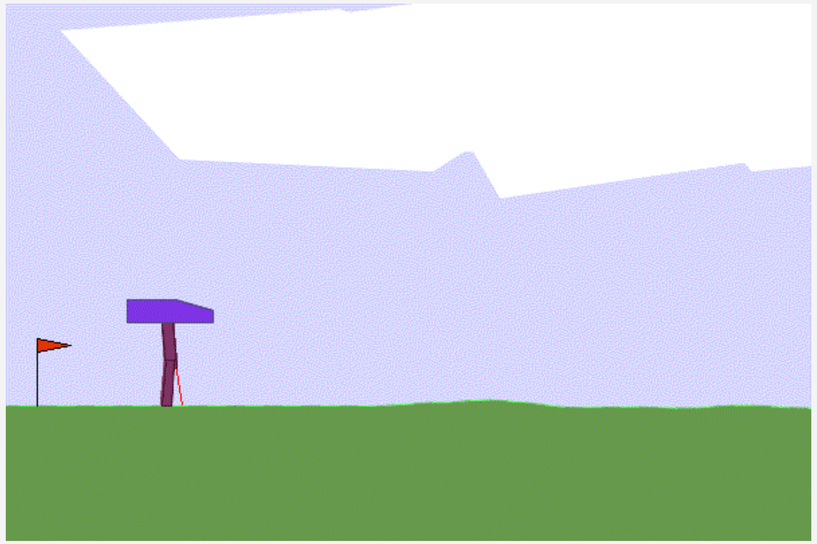
Mountain car



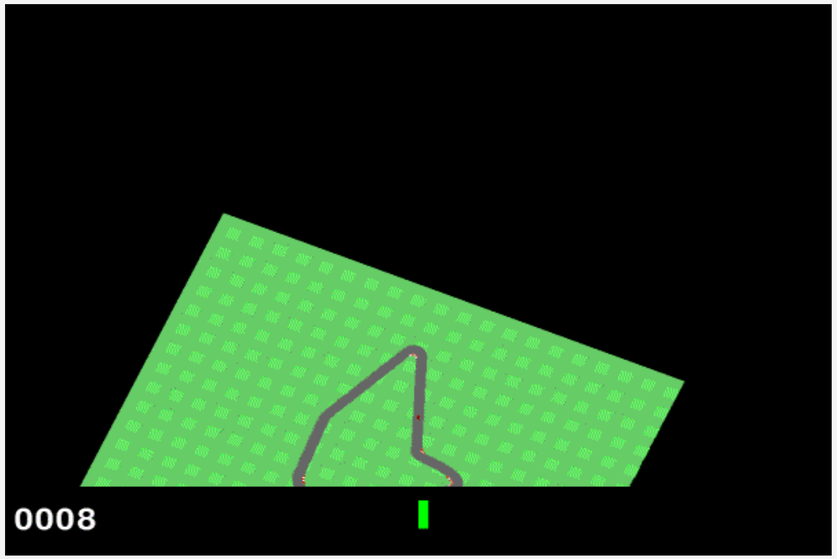
Pendulum

Gym: Box 2D

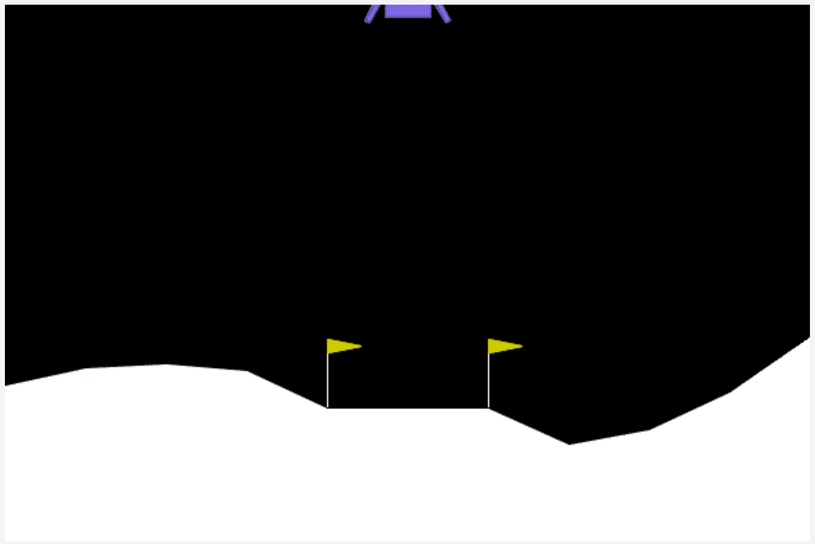
Problemas que involucran control físico



Bipedal walker



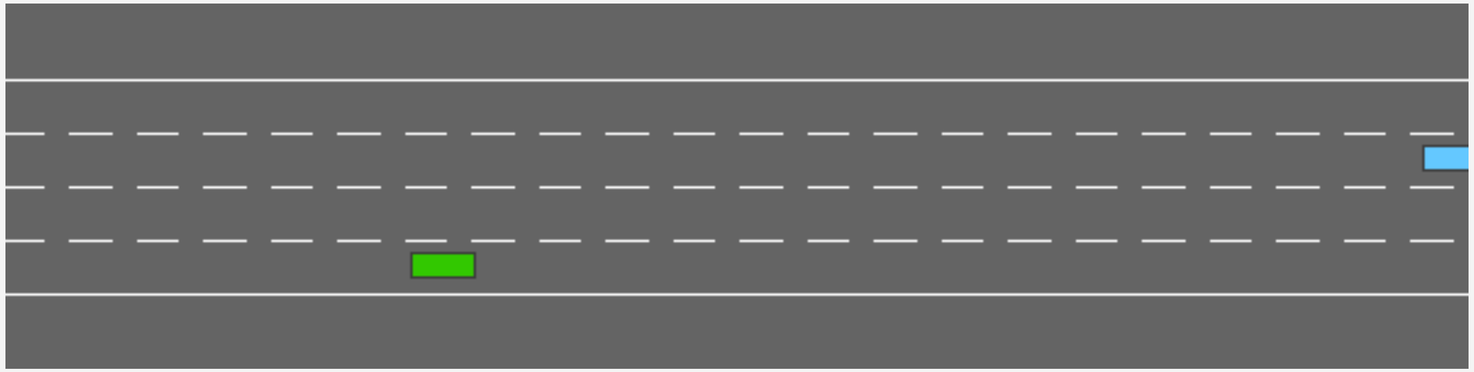
Car racing



Lunar lander

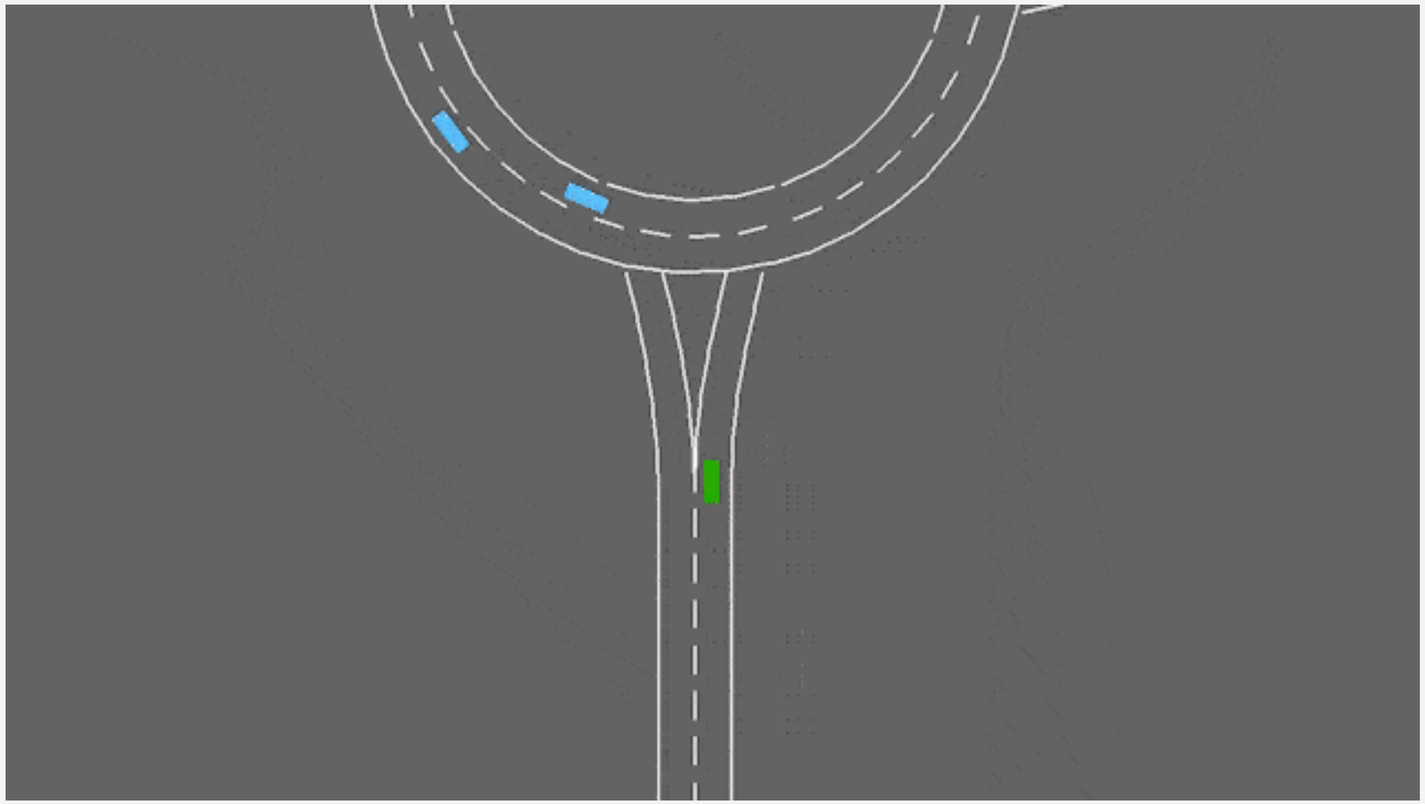
Gym: Terceras partes

Muchos más entornos. Por ejemplo, highway-env:



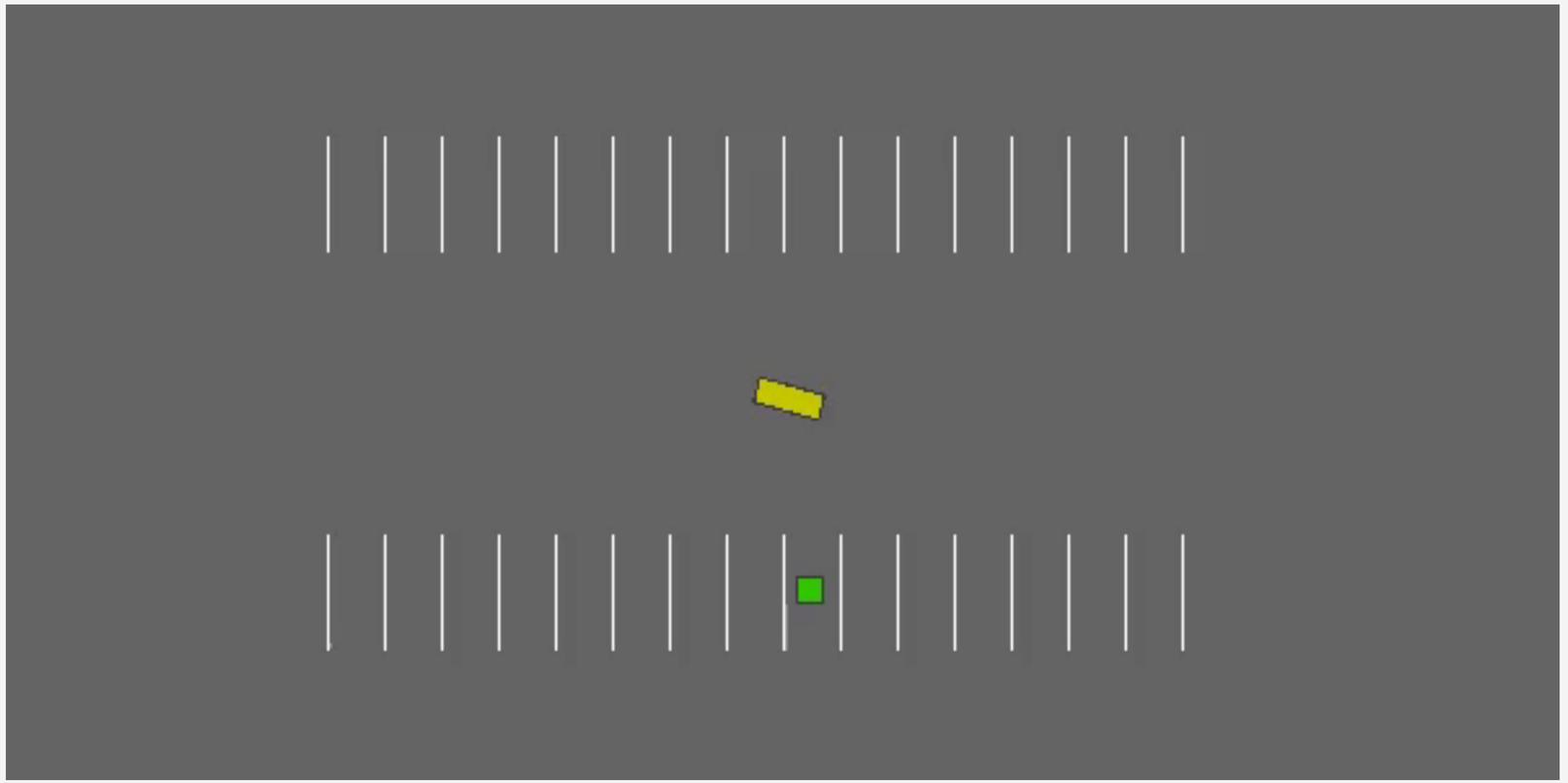
Gym: Terceras partes

Muchos más entornos. Por ejemplo, highway-env:



Gym: Terceras partes

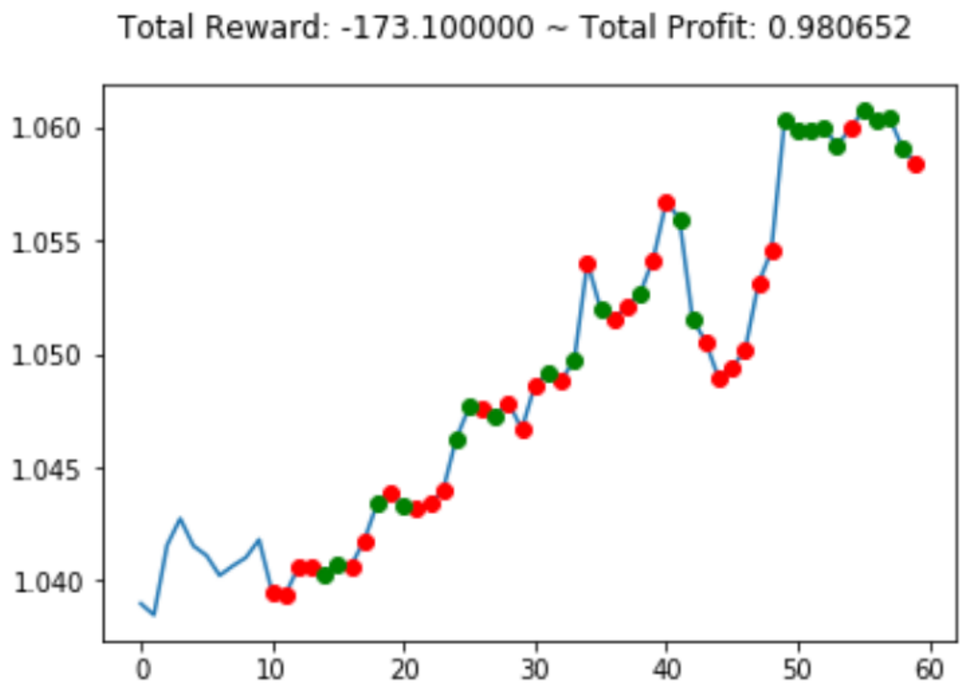
Muchos más entornos. Por ejemplo, highway-env:



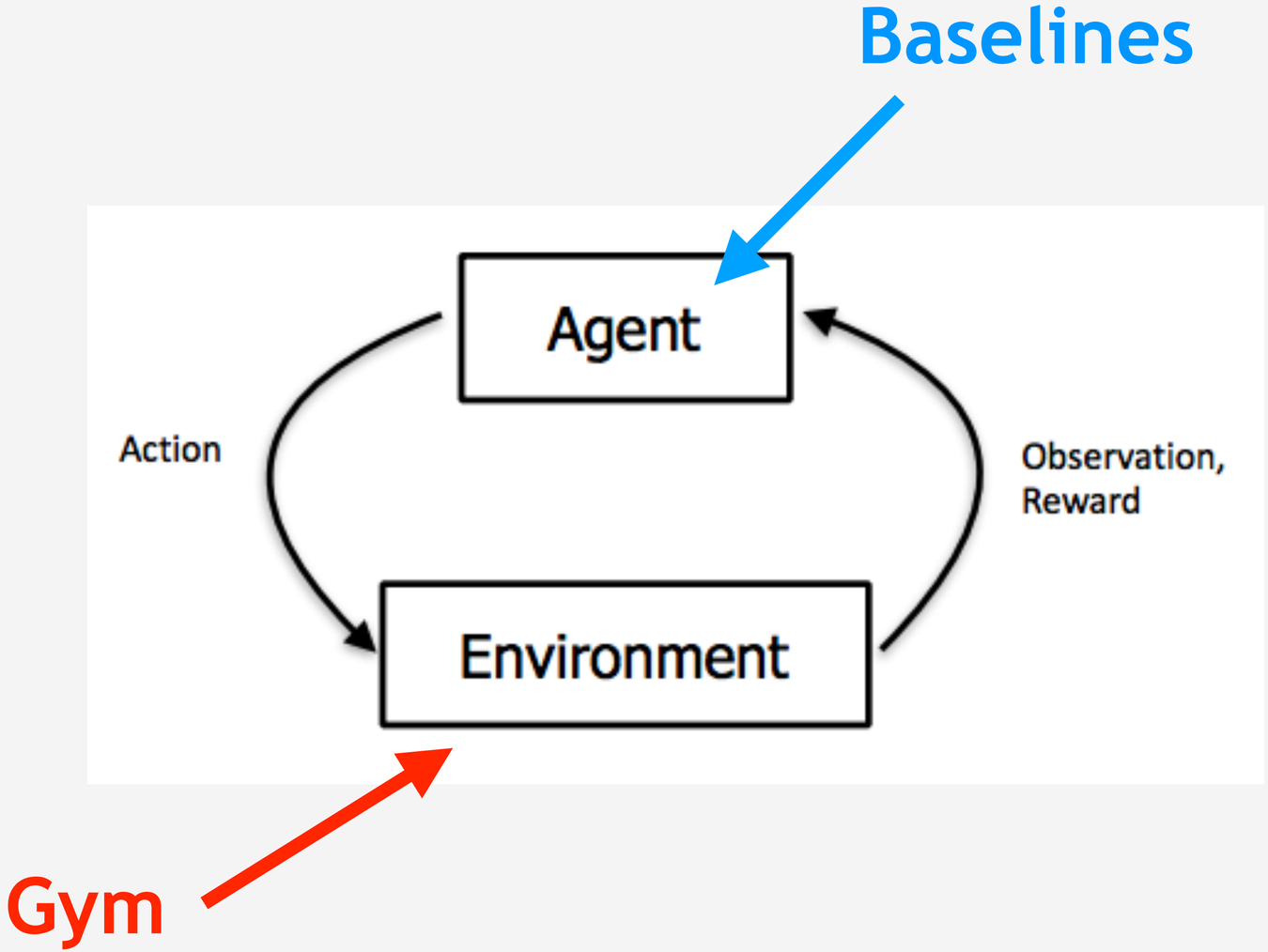
Gym: Terceras partes

Muchos más entornos. Por ejemplo, anytrading:

```
info: {'total_reward': -173.10000000000602, 'total_profit': 0.980652456904312, 'position': 0}
```



Gym y Baselines



OpenAI Baselines

https://github.com/openai/baselines

Baselines



OpenAI Baselines is a set of high-quality implementations of reinforcement learning algorithms.

These algorithms will make it easier for the research community to replicate, refine, and identify new ideas, and will create good baselines to build research on top of. Our DQN implementation and its variants are roughly on par with the scores in published papers. We expect they will be used as a base around which new ideas can be added, and as a tool for comparing a new approach against existing ones.

Stable Baselines

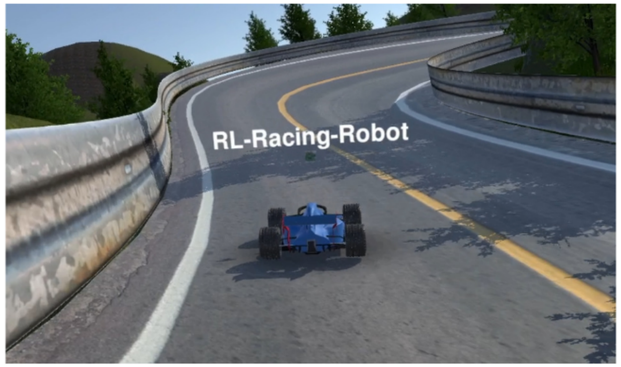
https://stable-baselines3.readthedocs.io/en/master/

Zoo

https://github.com/DLR-RM/rl-baselines3-zoo

RL Baselines3 Zoo: A Training Framework for Stable Baselines3 Reinforcement Learning Agents

RL Baselines3 Zoo is a training framework for Reinforcement Learning (RL), using Stable Baselines3.



It provides scripts for training, evaluating agents, tuning hyperparameters, plotting results and recording videos.

In addition, it includes a collection of tuned hyperparameters for common environments and RL algorithms, and agents trained with those settings.

We are **looking for contributors** to complete the collection!

Goals of this repository:

1. Provide a simple interface to train and enjoy RL agents
2. Benchmark the different Reinforcement Learning algorithms
3. Provide tuned hyperparameters for each environment and RL algorithm
4. Have fun with the trained agents!

This is the SB3 version of the original SB2 rl-zoo.